MAGIQ Installation Instructions

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This document will provide instructions for the installation of MAGIQ for both Windows and Mac computers. The instructions were originally developed for Windows users, but I adapted the instructions to allow Mac users to also download the programs.

These instructions took a large amount of time trying to find the optimal python version to support all MAGIQ dependencies. There are two MAGIQ dependencies that are missing from the instructions. The first is the MATLAB instructions, which is required for barstool rodent version. Dickson Wong's github has instructions on this installation that function as of today. The second missing dependency is pygamma. This is required for PINTS. It is a folder that is no longer accessible via the internet. To collect this folder, search through the lab computer and copy the pygamma folder to your own device.

All credit for MAGIQ goes to Dickson Wong and his collaborators. See here for the original (outdated) installation instructions, and a detailed outline of each of MAGIQ's functions.

Instructions for Windows Users

1. Download MAGIQ from this GitHub fork. Take note of it's location.
2. Download miniconda from here
    a. Miniconda can be found as "Anaconda Prompt" in your search bar after download.
3. In miniconda, type:
    a. *conda create -n python38 python=3.8.0*

       This creates a virtual environment that allows python 3.8 to run, even though it is technically outdated. You MUST use this version of python for all of the packages to be downloadable. This was the point in time where every package existed at the same time.

    b. To launch this virtual environment, every time you load up Miniconda, type:

       Conda activate python37

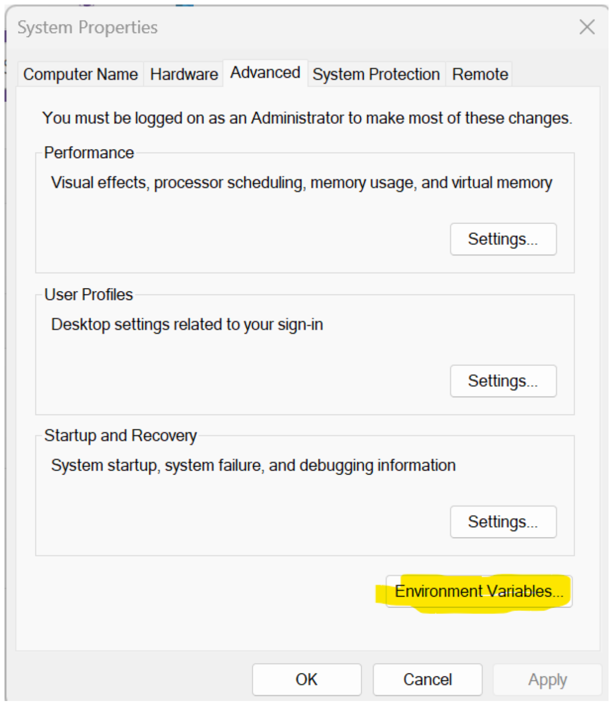
4. In the new python38 environment, you must now download the necessary packages. Type:
    a. *pip install PyQt5*
    b. *pip install scipy*
    c. *pip install matplotlib*
    d. *pip install pyfftw*
    e. *pip install future*

5. IDL download
    a. IDL is required to run fitman, as it was originally created here. Using this link, create an account, and register to be able to access the downloads. From there you must download the latest version of IDL. ENSURE YOU TAKE NOTE OF THE LOCATION OF THE DOWNLOAD (the folder path). You will use this later. Do not worry about licensing, ignore the license manager that will

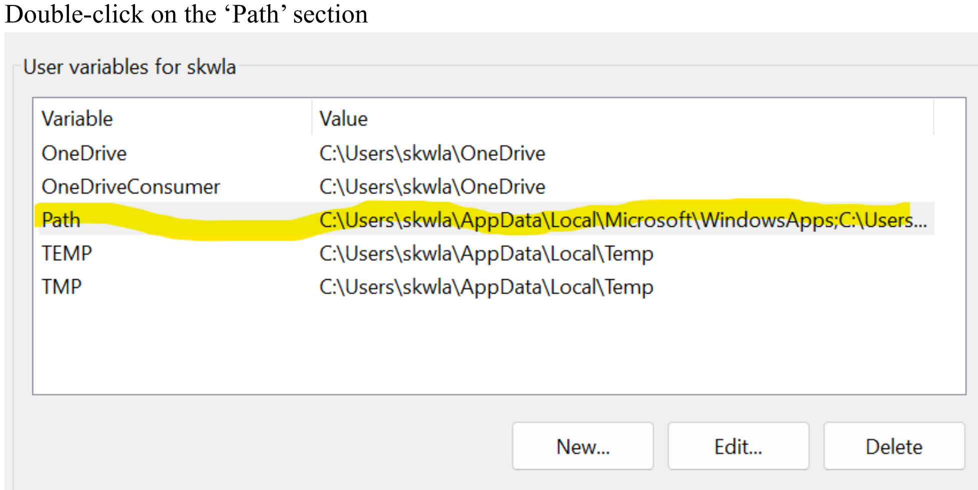
automatically open, you are using the virtual machine, which DOES NOT require a license. After downloading, restart your computer.

b. After restarting, search ‘environment variables’ in the Windows search bar, and open the control panel option.

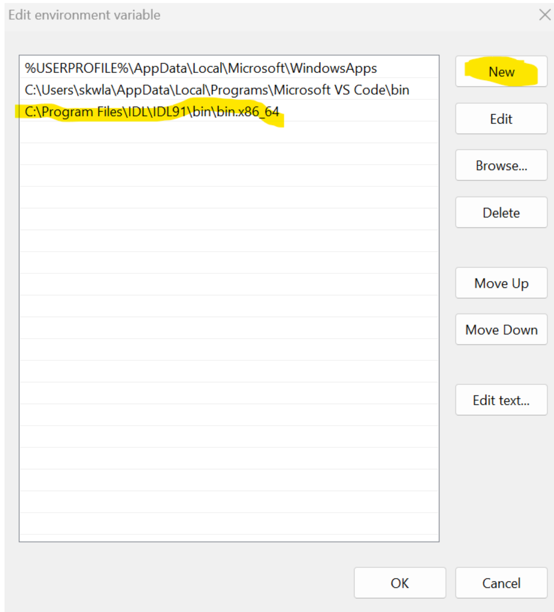
c. Select the “Environment Variables” button at the bottom



d. Double-click on the ‘Path’ section



e. Select ‘New” and input the path to the newly downloaded application, along with the path to the folder “bin.x86_64”. Search your computers files until you get there, it should look something like what is shown in my image. After inputting the path, add it.



You will need to update the path depending on it's location on your device, along with the IDL version.

You should now have a functioning version of MAGIQ on your device, with the exception of barstoolrv and PINTS.

To launch MAGIQ:

1. Open miniconda (may be labelled 'Anaconda Prompt' in your Windows search bar)
2. Type:
   a. Conda activate python38
3. Set the directory to your MAGIQ installation location. To do this, type cd /MAGIQ/PATH. I type *cd C:/users/skwla/MAGIQ*. Yours will be unique to where you downloaded MAGIQ to.
4. Type:
   a. *Python main.py*

MAGIQ should then open.

FSL Installation

Though not directly a part of MAGIQ, FSL will be needed for MRS analysis. Instructions for Windows download can be found here. You can also ask around the lab, as everyone likely has downloaded it and can help.

Instructions for Mac Users

These installation instructions will be slightly less detailed than that for Windows, since I used a Windows laptop, and created these instructions within a few hours on a Mac.

1. Download MAGIQ from this GitHub fork. Take note of it's location.
2. Download X-Quartz from the internet or app store.
3. Download miniconda from here
4. In miniconda, type:
    a. *conda create -n python38 python=3.8*

       This creates a virtual environment that allows python 3.8 to run, even though it is technically outdated. You MUST use this version of python for all of the packages to be downloadable. This was the point in time where every package existed at the same time.

    b. To launch this virtual environment, every time you load up Miniconda, type:

       Conda activate python37

5. In the new python38 environment, you must now download the necessary packages. Type:
    a. *pip install PyQt5*
    b. *pip install scipy*
    c. *pip install matplotlib*
    d. *conda install -c conda-forge pyfftw*
    e. *pip install future*

6. IDL download
    a. IDL is required to run fitman, as it was originally created here. Using this link, create an account, and register to be able to access the downloads. From there you must download the latest version of IDL. ENSURE YOU TAKE NOTE OF THE LOCATION OF THE DOWNLOAD (the folder path). You will use this later. Do not worry about licensing, ignore the license manager that will automatically open, you are using the virtual machine, which DOES NOT require a license. After downloading, restart your computer.
    b. You must now set IDL in the PATH. To do this, take remember the location and path that it was downloaded. Search through the IDL folder to find the subfolder named 'bin'. Then type:
        i. *path+= ('/path/to/IDL/bin')*
        ii. *export PATH*
    c. To verify this, type *echo $PATH* and look to see if the bin subfolder appears. If it does, everything should work. An example of how I did it on a Mac is below:
        i. *path+=('/Users/victorialittle/Applications/NV5/idl91/bin')*
        ii. *export PATH*
        iii. *echo $PATH*

You should now have a functioning version of MAGIQ on your device, with the exception of barstoolrv and PINTS.

To launch MAGIQ:

1. Open miniconda
2. Type:
    a. Conda activate python38

3. Set the directory to your MAGIQ installation location. To do this, type cd /MAGIQ/PATH. I type *cd C:/users/skwla/MAGIQ*. Yours will be unique to where you downloaded MAGIQ to.
4. Type:
    a. *Python main.py*
5. You may need to redo the PATH step (*path+= ('/path/to/IDL/bin'); export PATH*) if you are having issues. If the above does not work, retry after redoing the path step. You may need to do this every time.

MAGIQ should then open.